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RETAIL ANALYTICS & AI-POWERED SALES FORECASTING SYSTEM

1. Introduction

Modern retail businesses generate vast amounts of data across multiple stores, products, and customer interactions. Leveraging this data effectively can help organizations improve decision-making, optimize inventory, and increase revenue.

In this project, an AI-powered Retail Analytics and Forecasting System has been developed to analyze historical sales data, identify patterns, forecast future sales, and provide actionable business recommendations. The system is designed to support management in making data-driven decisions.

2. Dataset Description

The dataset used in this project contains retail sales transactions from January 2023 to December 2024, consisting of approximately 73,000 records.

Key Features:

- **Date** – Transaction date
- **Store_ID** – Unique store identifier
- **Product_Category & Subcategory** – Product classification
- **Unit_Price & Units_Sold** – Pricing and demand
- **Total_Sales & Revenue** – Sales metrics
- **Discount_Percentage** – Discount applied
- **Customer_Type** – New/Returning
- **Region** – Geographic segmentation
- **Store_Rating** – Performance indicator

The dataset is clean with no missing values, enabling efficient analysis and modeling.

3. Data Analysis & Key Insights

Exploratory Data Analysis (EDA) was performed to understand trends and patterns in the data.

Key Findings:

- Sales exhibit **consistent performance over time** with noticeable fluctuations due to dynamic demand.
- **March and August** show peak sales, indicating moderate seasonal patterns.
- Product categories contribute **almost equally to total revenue**, ensuring a balanced business model.

- No extreme dependency on a single category, reducing business risk.

4. Forecasting Models

To predict future sales, two approaches were used:

4.1 Machine Learning Model (Random Forest)

A Random Forest Regressor was implemented to predict sales using features such as pricing, discounts, stock levels, and time-based variables.

Performance:

- **MAE:** ~656
- **RMSE:** ~1039

Interpretation:

The model shows strong predictive performance with minimal error, indicating reliable forecasting capability.

4.2 Time-Series Model (ARIMA)

An ARIMA model was applied to capture temporal dependencies and forecast future sales trends.

Result:

- Forecast indicates **stable future sales patterns**
- No major fluctuations expected

5. Feature Importance Analysis

Feature importance analysis revealed key drivers of sales:

- **Units Sold**
- **Unit Price**
- **Discount Percentage**

6. Store Segmentation

K-Means clustering was used to segment stores into three categories:

- **High-performing stores**
- **Medium-performing stores**
- **Low-performing stores**

7. Business Recommendations

Based on the analysis, the following recommendations are provided:

- Increase inventory and marketing during peak months (March & August)
- Focus on improving low-performing stores using targeted promotions

- Optimize discount strategies to balance revenue and profitability
- Leverage weekend and seasonal demand through campaigns
- Maintain balanced investment across product categories

8. Dashboard (Power BI)

An interactive dashboard was developed using Power BI to visualize key insights.

Features:

- Monthly sales trends
- Category-wise performance
- Store performance analysis
- Regional sales distribution
- Interactive filters for dynamic exploration

9. Conclusion

This project successfully delivers an end-to-end Retail Analytics and Forecasting System that integrates data analysis, machine learning, and time-series forecasting.

The system provides:

- Actionable business insights
- Accurate sales predictions
- Store performance segmentation
- Interactive visualization